

1. The points (5, 2, 4), (6, -1, 2) and (8, -7, k) are collinear, if k is equal to
(a) -2 (b) 2 (c) 3 (d) -1
2. The angle between the lines $r = (4i - j) + s(2i + j - 3k)$ and $r = (i - j + 2k) + t(i - 3j + 2k)$ is
(a) $\frac{3\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{6}$
3. The co-ordinates of the foot of the perpendicular drawn from the point A(1, 0, 3) to the join of the points B(4, 7, 1) and C(3, 5, 3) are
(a) (5/3, 7/3, 17/3) (b) (5, 7, 17)
(c) (5/3, -7/3, 17/3) (d) (-5/3, 7/3, -17/3)
4. If A (1, 2, 3), B(6, 7, 8) C(1, 2, 5) and D (3, 0, 4) are given points, then the projection of $\vec{AB}$ on $\vec{CD}$ is
(a) 1/3 (b) 4/3 (c) 25/3 (d) 5/3
5. A Plane $x + 2y - 3z = 12$ has point P which is at minimum distance from the line joining A(1, 0, -3) and B(2, 3, -1) then AP^2 is equal to -
(a) 0 (b) 14 (c) 28 (d) 56
6. The tangent of the angle between a diagonal of a cube and the diagonal of a face (which meets the former) is-
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{3}$ (d) $\sqrt{\frac{2}{3}}$
7. If the points (0, 1, 2), (2, -1, 3) and (1, -3, 1) are the vertices of a triangle, then the triangle is
(a) Right angled (b) Isosceles right angled
(c) Equilateral (d) None of these
8. The point dividing the line joining the points (1, 2, 3) and (3, -5, 6) in the ratio 3:-5 is
(a) $\left(2, \frac{-25}{2}, \frac{3}{2}\right)$ (b) $\left(-2, \frac{25}{2}, \frac{-3}{2}\right)$
(c) $\left(2, \frac{25}{2}, \frac{3}{2}\right)$ (d) None of these
9. Points (1, 1, 1), (-2, 4, 1), (-1, 5, 5) and (2, 2, 5) are the vertices of a
(a) Rectangle (b) Square
(c) Parallelogram (d) Trapezium
10. Direction ratios of two lines are a, b, c and $\frac{1}{bc}, \frac{1}{ca}, \frac{1}{ab}$. The lines are
(a) Mutually perpendicular (b) Parallel
(c) Coincident (d) None of these
11. The co-ordinates of the foot of perpendicular drawn from the origin to the line joining the points (-9, 4, 5) and (10, 0, -1) will be
(a) (-3, 2, 1) (b) (1, 2, 2)
(c) (4, 5, 3) (d) None of these
12. Distance of the point (2, 3, 4) from the plane $3x - 6y + 2z + 11 = 0$ is
(a) 1 (b) 2 (c) 3 (d) 0
13. The equation of the perpendicular from the point (α, β, γ) to the plane $ax + by + cz + d = 0$ is
(a) $a(x - \alpha) + b(y - \beta) + c(z - \gamma) = 0$
(b) $\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c}$
(c) $a(x - \alpha) + b(y - \beta) + c(z - \gamma) = abc$
(d) None of these
14. The length and foot of the perpendicular from the point (7, 14, 5) to the plane $2x + 4y - z = 2$, are
(a) $\sqrt{21}, (1, 2, 8)$ (b) $3\sqrt{21}, (3, 2, 8)$
(c) $21\sqrt{3}, (1, 2, 8)$ (d) $3\sqrt{21}, (1, 2, 8)$

15. A plane meets the co-ordinate axes in A, B, C and (α, β, γ) is the centered of the triangle ABC . Then the equation of the plane is
- (a) $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3$ (b) $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$
- (c) $\frac{3x}{\alpha} + \frac{3y}{\beta} + \frac{3z}{\gamma} = 1$ (d) $\alpha x + \beta y + \gamma z = 1$
16. If the length of perpendicular drawn from origin on a plane is 7 units and its direction ratios are -3, 2, 6, then that plane is
- (a) $-3x + 2y + 6z - 7 = 0$ (b) $-3x + 2y + 6z - 49 = 0$
- (c) $3x - 2y + 6z + 7 = 0$ (d) $-3x + 2y - 6z - 49 = 0$
17. In a three dimensional xyz space the equation $x^2 - 5x + 6 = 0$ represents
- (a) Points (b) Plane
- (c) Curves (d) Pair of straight line
18. If the points $(1, 1, k)$ and $(-3, 0, 1)$ be equidistant from the plane $3x + 4y - 12z + 13 = 0$, then $k =$
- (a) 0 (b) 1 (c) 2 (d) None of these
19. The equation of the plane which bisects the line joining the points $(-1, 2, 3)$ and $(3, -5, 6)$ at right angle, is
- (a) $4x - 7y - 3z = 8$ (b) $4x + 2y - 3z = 28$
- (c) $4x - 7y + 3z = 28$ (d) $4x - 7y - 3z = 28$
20. The distance between the line $\frac{x-1}{3} = \frac{y+2}{-2} = \frac{z-1}{2}$ and the plane $2x + 2y - z = 6$ is
- (a) 9 (b) 1 (c) 2 (d) 3